Practice Set A

Class 12 Mathematics, Chapter 10: Vector Algebra

Maximum marks: **47** **20** questions

Name _____ Date _____ Score _____ / 47

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1 — 8 marks

1 MCQ, PYQ-style — 1 mark

The value(s) of λ for which $\lambda(2\hat{i} - \hat{j} + 2\hat{k})$ is a unit vector is/are:

(A) $\pm\frac{1}{3}$ **(B)** ± 3 **(C)** $\frac{1}{3}$ only **(D)** $\pm\frac{1}{9}$

2 MCQ, CBSE SQP 2024-25 — 1 mark

If $|\vec{a}| = 3$, $|\vec{b}| = 4$ and $|\vec{a} + \vec{b}| = 5$, then $|\vec{a} - \vec{b}| =$

(A) 3 **(B)** 4 **(C)** 5 **(D)** 8

3 MCQ, CBSE SQP 2024-25 — 1 mark

The value of α for which the angle between $\vec{p} = 2\alpha^2\hat{i} - 3\alpha\hat{j} + \hat{k}$ and $\vec{q} = \hat{i} + \hat{j} + \alpha\hat{k}$ is obtuse, is:

(A) $\mathbb{R} - [0, 1]$ **(B)** $(0, 1)$ **(C)** $[0, \infty)$ **(D)** $[1, \infty)$

4 MCQ, PYQ-style — 1 mark

If $|\vec{a} \times \vec{b}|^2 + |\vec{a} \cdot \vec{b}|^2 = 400$ and $|\vec{a}| = 5$, then $|\vec{b}|$ is equal to:

(A) 4 **(B)** 16 **(C)** 80 **(D)** 400

5 MCQ, CBSE SQP 2025-26 — 1 mark

A bird flies through a distance in a straight line given by the vector $\hat{i} + 2\hat{j} + \hat{k}$. A man standing beside a straight metro rail track given by $\vec{r} = (3 + \lambda)\hat{i} + (2\lambda - 1)\hat{j} + 3\lambda\hat{k}$ is observing the bird. The projected length of its flight on the metro track is:

(A) $\frac{6}{\sqrt{14}}$ units **(B)** $\frac{14}{\sqrt{6}}$ units **(C)** $\frac{8}{\sqrt{14}}$ units **(D)** $\frac{5}{\sqrt{6}}$ units

Q No.	Question	Marks
6	Assertion-Reason, CBSE SQP 2025-26 **Assertion (A):** Given two non-zero vectors $\vec{a}$ and $\vec{b}$. If $\vec{r}$ is another non-zero vector such that $\vec{r} \times (\vec{a} + \vec{b}) = \vec{0}$, then $\vec{r}$ is perpendicular to $\vec{a} \times \vec{b}$. **Reason (R):** The vector $(\vec{a} + \vec{b})$ is perpendicular to the plane of $\vec{a}$ and $\vec{b}$. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, CBSE SQP 2025-26 The two vectors $\hat{i} + \hat{j} + \hat{k}$ and $3\hat{i} - \hat{j} + 3\hat{k}$ represent the two sides OA and OB, respectively of a $\triangle OAB$, where O is the origin. The point P lies on AB such that OP is a median. Find the area of the parallelogram formed by the two adjacent sides as OA and OP.	2 marks

Level 2

16 marks

8	Short answer, PYQ-style Find a vector of magnitude 5 units in the direction of the resultant of the vectors $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$.	2 marks
9	Short answer, PYQ-style Find a unit vector perpendicular to the plane ABC, where A, B and C are the points $(3, -1, 2)$, $(1, -1, -3)$ and $(4, -3, 1)$ respectively.	2 marks
10	Short answer, CBSE SQP 2024-25 A person standing at $O(0, 0, 0)$ is watching an aeroplane which is at the coordinate point $A(4, 0, 3)$. At the same time he saw a bird at the coordinate point $B(0, 0, 1)$. Find the angles which $\overrightarrow{BA}$ makes with the x, y and z axes.	2 marks
11	Short answer, PYQ-style Express the vector $\vec{a} = 5\hat{i} - 2\hat{j} + 5\hat{k}$ as the sum of two vectors such that one is parallel to the vector $\vec{b} = 3\hat{i} + \hat{k}$ and the other is perpendicular to $\vec{b}$.	3 marks
12	Short answer, CBSE SQP 2024-25 An ant is moving along the vector $\vec{l_1} = \hat{i} - 2\hat{j} + 3\hat{k}$. Few sugar crystals are kept along the vector $\vec{l_2} = 3\hat{i} - 2\hat{j} + \hat{k}$ which is inclined at an angle θ with the vector $\vec{l_1}$. Then find the angle θ. Also find the scalar projection of $\vec{l_1}$ on $\vec{l_2}$.	3 marks
13	Short answer, PYQ-style If $\lvert\vec{a}\rvert = 3$ and $\lvert\vec{b}\rvert = 4$, find the value(s) of λ for which the vectors $\vec{a} + \lambda\vec{b}$ and $\vec{a} - \lambda\vec{b}$ are perpendicular.	2 marks

Q No.	Question	Marks
14	Short answer, PYQ-style During a cricket match, the positions of the bowler, the wicket-keeper and the leg-slip fielder are in a line, given by $\vec{B} = 2\hat{i} + 8\hat{j}$, $\vec{W} = 6\hat{i} + 12\hat{j}$ and $\vec{F} = 12\hat{i} + 18\hat{j}$ respectively. Find the ratio in which the wicket-keeper divides the line segment joining the bowler and the fielder.	2 marks

Level 3 — 23 marks

15	Short answer, PYQ-style If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are mutually perpendicular vectors of equal magnitudes, show that the vector $\vec{a} + \vec{b} + \vec{c}$ is equally inclined to $\vec{a}$, $\vec{b}$ and $\vec{c}$.	3 marks
16	Long answer, PYQ-style Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{j} - \hat{k}$. Find a vector $\vec{c}$ such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$.	5 marks
17	Short answer, PYQ-style The scalar product of the vector $\hat{i} + \hat{j} + \hat{k}$ with a unit vector along the sum of the vectors $2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\lambda\hat{i} + 2\hat{j} + 3\hat{k}$ is equal to one. Find the value of λ.	3 marks
18	Short answer, PYQ-style Using vectors, find the value of k such that the points $A(k, -10, 3)$, $B(1, -1, 3)$ and $C(3, 5, 3)$ are collinear.	3 marks
19	Long answer, Board-pattern Using vectors, show that the points $A(1, 1, 1)$, $B(2, 3, 3)$ and $C(3, 2, -1)$ are the vertices of an isosceles right-angled triangle. Hence find the area of the triangle and the measure of $\angle B$.	5 marks
20	Case study, Board-pattern **Case Study:** The triangular sail of a toy boat is a flat piece of cloth with corners O, P and Q. Taking O as the origin, $\overrightarrow{OP} = 2\hat{i} + 2\hat{j} + \hat{k}$ and $\overrightarrow{OQ} = 3\hat{i} + 4\hat{k}$ (units in metres). **(i)** Find the length of the edge OP and the unit vector along $\overrightarrow{OP}$. — 1 mark **(ii)** Find the projection of $\overrightarrow{OQ}$ on $\overrightarrow{OP}$. — 1 mark **(iii)** Find the area of the sail. — 2 marks	4 marks

Answer key

Chapter 10: Vector Algebra, Practice Set A. Full step-wise solutions are in the solutions sheet.

Level 1

1	(A) $\pm\frac{1}{3}$
2	(C) 5
3	(B) $(0, 1)$
4	(A) 4
5	(C) $\frac{8}{\sqrt{14}}$ units
6	(C) A is true, but R is false
7	Area $= 2\sqrt{2}$ sq units

Level 2

8	$\frac{3\sqrt{10}}{2}\hat{i} + \frac{\sqrt{10}}{2}\hat{j}$
9	$\pm\frac{1}{\sqrt{165}}(-10\hat{i} - 7\hat{j} + 4\hat{k})$
10	$\cos^{-1}\frac{2}{\sqrt{5}}$, $\frac{\pi}{2}$, $\cos^{-1}\frac{1}{\sqrt{5}}$
11	$\vec{a} = (6\hat{i} + 2\hat{k}) + (-\hat{i} - 2\hat{j} + 3\hat{k})$
12	$\theta = \cos^{-1}\frac{5}{7}$; projection $= \frac{10}{\sqrt{14}} = \frac{5\sqrt{14}}{7}$
13	$\lambda = \pm\frac{3}{4}$
14	$2 : 3$ internally

Level 3

15	Each angle is $\cos^{-1}\frac{1}{\sqrt{3}}$, so $\vec{a} + \vec{b} + \vec{c}$ is equally inclined to $\vec{a}, \vec{b}, \vec{c}$.
16	$\vec{c} = \frac{5}{3}\hat{i} + \frac{2}{3}\hat{j} + \frac{2}{3}\hat{k} = \frac{1}{3}(5\hat{i} + 2\hat{j} + 2\hat{k})$
17	$\lambda = 1$
18	$k = -2$
19	Right angle at A, $AB = AC = 3$; area $= \frac{9}{2}$ sq units; $\angle B = \frac{\pi}{4}$
20	(i) 3 m, $\frac{1}{3}(2\hat{i} + 2\hat{j} + \hat{k})$; (ii) $\frac{10}{3}$ m; (iii) $\frac{5\sqrt{5}}{2}$ sq m